# Nathan J. Aronson

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## Education

**University of Pennsylvania** **Aug. 2024 – May 2028**

*B.S.E. in Computer Science (Systems Concentration), Mathematics Minor* *Philadelphia, PA*
*M.S.E. in Computer Science (Artificial Intelligence Concentration)*
GPA: 3.95/4.00

**Relevant Coursework:** Operating Systems (Graduate), Databases (Graduate), Machine Learning (Graduate), Big Data Analytics (Graduate), Advanced Algorithms, Data Structures & Algorithms, Computer Systems, Probability, Linear Algebra

## Technical Skills

**Languages**: C++, Python, Java, Rust, C, OCaml, SQL, Swift, JavaScript/TypeScript
**Libraries & Frameworks**: NumPy, Pandas, PyTorch, POSIX API, Tokio, FastAPI, React.js, Node.js, SwiftUI
**Technologies**: Git, Docker, Linux, PostgreSQL, MongoDB, Neo4j, Modal, OpenClaw, Claude Code

## Experience

**Arcesium** **June 2026 – Present**

*Software Engineer Intern* *New York, NY*

- Build a crypto accounting engine rebuilding positions and P&L across 3 chains and 10 position types; surface 2.75% execution price improvement via MEV-protected routing; independently reconcile with the fund manager's book.
- Spearhead retry mechanism for cross-platform trade synchronization that automatically resubmits rejected transactions.

**Astera Holdings** **June 2025 – Aug. 2025**

*Lead Quantitative Engineer* *Remote*

- Led team of three to build a prediction market data acquisition service; developed Kalshi WebSocket API Rust crate.
- Built a real-time arbitrage detection tool matching Polymarket and Kalshi tickers by name similarity and historical price trends, used by the firm's quantitative researchers.

**Burns Engineering, Inc.** **May 2025 – Aug. 2025**

*Software Engineer Intern* *Pittsburgh, PA*

- Developed train safety system in Rust, integrated PCB interface board and keypad communication over Modbus RS485; designed a 64-bit protocol with 13 command/status signals and a multi-threaded Python keypad simulator.
- Built safety server file verification tool with Tkinter that replaced a 30 minute manual inspection process.

## Community Involvement

**Penn Labs** **Sept. 2025 – Present**

*iOS Mobile Engineer* *Philadelphia, PA*

- Maintain and extend production-grade mobile application used by 10,000+ Penn students, contributing to 12+ year-old repo with 3,000+ commits.
- Redesigned laundry tab, migrating 5,000 lines of legacy UIKit code to SwiftUI with alarms and live activities.

**University of Pennsylvania** **Jan. 2025 – Present**

*Head CIS 1200 Teaching Assistant* *Philadelphia, PA*

- Lead staff of 30+ TAs for a 300+ student course, running weekly staff meetings and coordinating with professors.
- Teach a weekly recitation, write exam questions, grade assignments, and hold office hours on functional and object-oriented programming in OCaml and Java.

## Projects

**Darwin – 2nd Place Winner Cubist Hackathon Spring 2026** | *React, FastAPI, Modal, Tailwind, Claude/Gemini Agents*

- Designed an evolutionary chess engine where an orchestrating agent harnesses 16 worker agents to propose and implement mutations, competing to become the next base model.
- Parallelized round-robin tournaments across a network of Modal 40 sandboxes, yielding a 6x speedup per generation.

**PennOS** | *C, POSIX API*

- Developed a user-level UNIX-like operating system from scratch in C, including a round-robin scheduler with context switching, PCBs, and zombie/orphan handling, and a FAT-16 file system.
- Implemented command-line shell with pipelines, job control, foreground/background execution, I/O redirection, and signal handling.

**Arc – Programming Language** | *Rust, Java, WebAssembly*

- Implemented Pratt parser, single-pass bytecode compiler, and stack-based VM in Rust supporting variables, conditionals, loops, and closures; compiled to WebAssembly to run in an interactive web playground.
- Achieved 7x speedup over tree-walking Java implementation via designing bytecode and optimized static rules table.